

Agentic RAG with a self-improving retriever

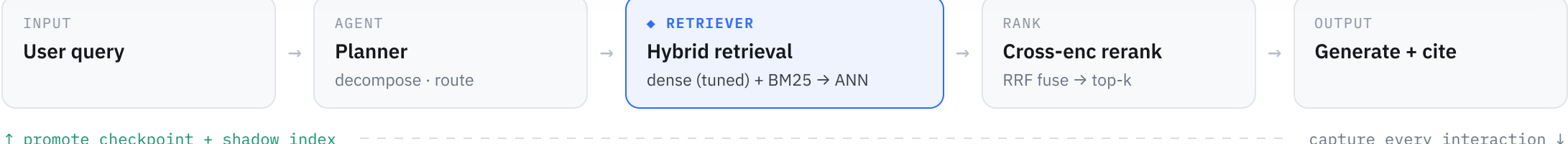
A closed-loop system: an **agent plans multi-step retrieval** at query time; every production interaction is harvested into training signal; and the dense **retriever is continually fine-tuned** on mined hard negatives — with debiasing and guardrails that keep the feedback loop from collapsing onto its own history.

SCALE ~500M docs · high QPS AUDIENCE ML / infra eng RETRIEVER dense bi-encoder + rerank LOSS InfoNCE + cross-encoder distill

01 · Orchestration 02 · Serving 03 · Feedback 04 · Pair mining 05 · Fine-tune 06 · Infra 07 · Evaluation 08 · Guardrails

THE CLOSED LOOP

SERVE · QUERY TIME



01 Agentic orchestration & retrieval planning

The agent turns one query into an adaptive retrieval program, not a single vector lookup.

A planner LLM decomposes the request into sub-queries and selects tools (search, filters, structured lookups). Each hop runs **hybrid retrieval** — a fine-tuned dense bi-encoder in parallel with BM25 — fused via **reciprocal-rank fusion**, then a cross-encoder reranks the top ~100 to top-k.

The agent inspects a **sufficiency check** (coverage, contradiction, confidence) and may reformulate, expand, or stop. This iterative behavior is the richest source of training signal: every reformulation is an implicit "the last retrieval missed" label.

Every step — sub-queries, candidate IDs, scores, chosen docs, the final answer, and grounding spans — is logged to a structured trace keyed by `trace_id` for later attribution.

LATENCY BUDGET · P99

Planner (small LLM)	~120 ms
Query encode (GPU)	~6 ms
ANN search (sharded)	~15 ms
Cross-enc rerank ×100	~40 ms
Retrieval subtotal	< 70 ms

note — rerank dominates; cap candidates and batch on GPU. Generation streams, so it's off the retrieval critical path.

02 Retriever model & high-QPS serving

Precompute what you can; the only query-time neural cost is one encode.

MODEL CHOICE

Bi-encoder base (BGE / GTE / E5-class, ~300M) distilled to a ~100M query encoder for latency. Document embeddings are precomputed offline, so model size only taxes the corpus re-embed job, not serving.

Where quality justifies the memory, a **ColBERT-style late-interaction** variant is the upgrade path — token-level MaxSim at the cost of a larger index. The cross-encoder reranker (a fine-tuned ~300M classifier) is where most head-precision comes from.

SERVING TOPOLOGY

- Query encoder** on Triton/TensorRT, GPU-batched, autoscaled by QPS.
- Vector index** sharded HNSW in-memory across replicas; IVF-PQ fallback when RAM-bound. Scatter-gather + merge.
- Caches**: exact + semantic query cache, hot-doc embedding cache.
- Atomic promotion**: encoder checkpoint and its matching index version flip together — never mix embedding spaces.

03 Production feedback & labeling

Four signal classes, fused into a per-(query, doc) relevance estimate with a propensity weight.

SIGNAL	WHAT IT CAPTURES	STRENGTH	BIAS RISK
👍 Explicit rating	User thumbs up/down on the answer	High intent, sparse	self-select
Implicit	Citation clicks, dwell, copy, no-reformulate	Dense, cheap	position
LLM-judge grounding	Is the answer supported by <i>this</i> doc?	Scalable, per-doc	model bias
Agent tool success	Did retrieval end the loop vs. force a re-query?	Free, causal-ish	low
Human queue	Annotators adjudicate low-confidence / conflicting cases	Gold, slow	low

attribution — a good answer doesn't credit every retrieved doc. Use citation grounding + an LLM attributor to assign credit only to docs whose spans actually appear in the answer. Uncited-but-retrieved docs from a good answer become **candidate hard negatives**, not positives.

04 Training-pair mining

Turn noisy traces into clean (query, positive, hard-negatives) tuples.

POSITIVES

- Docs cited in 👍 answers
- High LLM-judge grounding score
- Clicked + long-dwell citations
- Human-adjudicated relevant

HARD NEGATIVES

- High ANN rank but uncited in a good answer
- Retrieved in a 👎 / re-queried turn
- BM25 top that dense missed (and vice-versa)
- ANN-mined from the *current* checkpoint each epoch

Denoise before training: dedup near-identical docs, drop false negatives with an LLM/cross-encoder consistency filter (a "negative" the teacher scores highly is likely a missed positive), cap pairs-per-query to fight popularity skew, and attach an **inverse-propensity weight** so heavily-shown docs don't dominate the gradient.

05 The retriever fine-tuning loop

Contrastive objective + teacher distillation, with explicit anti-forgetting.

OBJECTIVE

InfoNCE / MNRL over in-batch + mined hard negatives, temperature-scaled.

Plus **cross-encoder distillation** (MarginMSE / KL to the reranker's scores) — the teacher densifies sparse production labels.

ANTI-FORGETTING

Replay buffer of historical + golden positives mixed into every batch.

KL / L2-SP anchor to the previous checkpoint bounds embedding drift so the index stays compatible-ish between rebuilds.

CADENCE

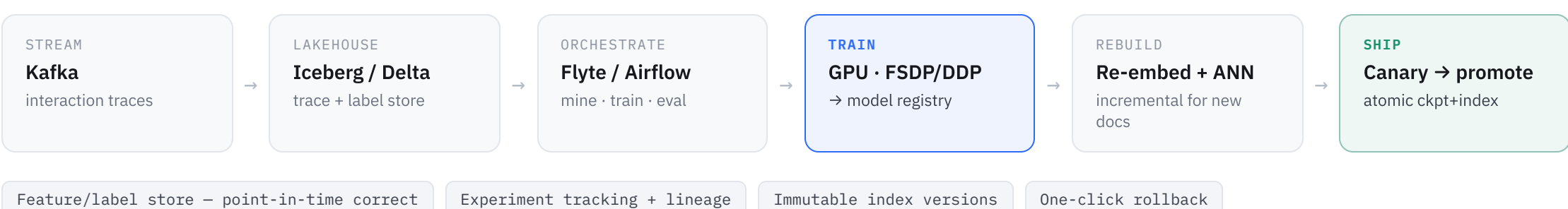
Nightly incremental fine-tune on the day's mined pairs → shadow index → offline gate.

Weekly full re-train + full corpus re-embed to reset accumulated drift.

```
# nightly loop
pairs = mine(traces[today], propensity_weighted=True)
pairs = denoise(pairs, teacher=cross_encoder) # drop false negs
neg = ann_hard_negatives(index=current_ckpt, k=15)
loss = infonce(q, pos, neg, tau=0.02) + lambda*marginmse(student, teacher) + beta*kl(student, prev_ckpt)
ckpt = fsdp_train(base=prev_ckpt, data=pairs+replay, epochs=1)
if offline_gate(ckpt) and canary_online(ckpt): promote(ckpt, rebuild_index=True)
```

06 Data & training infrastructure

Stream in, batch-materialize, train, rebuild, canary — every stage versioned.



07 Evaluation & promotion gates

A checkpoint ships only if it clears offline gates *and* a live canary — no exceptions.

OFFLINE · FROZEN GOLDEN SET	
nDCG@10 / MRR	ranking quality
Recall@k	candidate coverage
Golden regression	no drop vs. prod
Leakage audit	golden ≠ feedback

ONLINE · CANARY A/B + INTERLEAVING	
Answer correctness	judge + human
👍 rate · CTR	user signal
Task success	loop resolved
p99 latency · cost	guardrail

note — team-draft / interleaving compares old vs new retriever on the *same* traffic and needs far less exposure than a full A/B to detect a ranking win. Use it as the fast gate; A/B confirms downstream answer lift.

08 Guardrails & failure modes

A retriever trained on its own outputs will happily overfit to itself. Design against it.

FAILURE MODE	MITIGATION
Feedback-loop collapse	Retriever only ever sees / reinforces docs it already surfaces. → ε-exploration + diversity injection into candidates; log randomized slots to gather unbiased labels.
Position / presentation bias	Clicks favor top slots regardless of relevance. → IPS / propensity weighting : train a click model; prefer interleaving for eval.
LLM-judge label noise	Judge is systematically wrong on some slices. → calibrate against human gold; audit sampled labels; ensemble / abstain on low confidence.
Embedding drift vs. stale index	New encoder, old doc vectors → silent quality cliff. → atomic checkpoint+index promotion ; drift anchor in loss; compatibility check pre-flip.
Catastrophic forgetting	Nightly tuning erases rare-but-important knowledge. → replay buffer + golden set in every batch; weekly full retrain.
Distribution / query shift	Traffic mix moves under the model. → drift monitors on query + score distributions; auto-rollback on gate breach.
Adversarial / spam feedback	Coordinated 👍/👎 to promote content. → robust aggregation , per-user rate/trust weighting, anomaly detection on signal spikes.

